



ThinkStack

Testing Phase Document

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Chapter 1

TEST PLAN

1.1 Purpose and Scope

This document records how ThinkStack is tested, what the tests establish, and what they currently report. It covers the backend, the interface, the contract between them, and the packaged application as installed on a user's machine.

It does not yet cover cross-platform verification of the replaced interface or of the Scribe writing path. Section 5.3 states precisely what is outstanding and why, rather than leaving the omission implicit.

1.2 Testing Strategy

Four layers, each establishing something the layer below it cannot.

Table 1.1: Testing layers and what each establishes

Layer	Tool	What only this layer can establish
Unit (backend)	pytest	That a function computes what it claims, including its failure modes.
Unit (interface)	vitest, jsdom	That a component renders and behaves under interaction, without a browser.
Contract	pytest, source parsing	That both sides of the API agree, parsed from source rather than asserted by hand.
Reachability	live HTTP	That the handlers <i>run</i> . Parsing proves the paths match; only calling them proves anything executes.
Packaged application	validate_bundle.py	That the <i>installed</i> program works. A suite run from a source checkout cannot see what the freezer omitted.

Why the last two layers exist. Version 1.0.0 passed every test and every build, was published, and did not start: the frozen backend silently dropped the inference library's shared objects and the embedding model's data, because neither package ships a hook for the freezer. And when the interface was replaced, the contract test was found to be hardcoded to the old directory, the moment the replacement shipped it would have gone on passing while validating a tree nobody built.

1.3 Test Environment

Table 1.2: Test environment

Item	Value
Development machine	16 GB memory, 16-core x86-64 CPU, no usable accelerator
Operating system	Fedora Linux (kernel 7.1.3)
Python	3.14, dependencies pinned exactly
Node	22.x
Inference	llama.cpp via a local client; Qwen2.5 0.5B and 1.5B, Q4_K_M
Embeddings	all-MiniLM-L6-v2, local
Typesetting	Tectonic, bundled
Continuous integration	GitHub Actions, Ubuntu runners

Interface tests are run with file parallelism disabled. Vitest defaults to one worker per core; on a 16-core machine already running an editor that is sixteen Node processes each with its own DOM implementation, which exhausted memory and froze the machine. Single-threaded the same 198 tests complete in about five seconds.

Chapter 2

TEST CASES

2.1 Metadata Extraction

Table 2.1: Metadata extraction test cases

ID	Case	Expected	Result
ME-01	Largest horizontal text at the top of page 1	Exact title	Pass
ME-02	Rotated archive stamp larger than the title	Stamp ignored	Pass
ME-03	Horizontal margin furniture (journal banner)	Ignored	Pass
ME-04	Title wrapping to a second line	Both lines joined	Pass
ME-05	Same font reused lower on the page	Not joined to the title	Pass
ME-06	Small-capitals title (two sizes, one baseline)	Whole title	Pass
ME-07	Font change mid-word	No space inserted	Pass
ME-08	Ligature stored as one glyph	Expanded	Pass
ME-09	Accent stored before its letter	Recomposed	Pass
ME-10	Hyphenated title across a line break	De-hyphenated	Pass
ME-11	Authors in columns	Each cell a name	Pass
ME-12	Authors comma-separated in one cell	Split correctly	Pass
ME-13	One author per row with affiliation	Name kept, affiliation dropped	Pass
ME-14	Affiliation row naming only some employers	Whole row rejected	Pass
ME-15	Phrase repeated in the author region	Treated as an address	Pass
ME-16	Accented institution name	Still recognised	Pass
ME-17	Email in brace form	Not read as a name	Pass
ME-18	Scanned page, no text layer	Empty, not a guess	Pass

ID	Case	Expected	Result
ME-19	Year from archive identifier vs. stray number	Identifier wins	Pass
ME-20	Paper stating no year	Empty	Pass
ME-21	Implausible year	Refused	Pass
ME-22	Plausible title, no authors	Model fallback invoked	Pass
ME-23	Model returns an empty field	Good value retained	Pass
ME-24	Model disabled and result implausible	Returned unchanged	Pass

2.2 Documents, Storage and Encryption

Table 2.2: Document lifecycle and encryption test cases

ID	Case	Expected	Result
DS-01	Upload a valid PDF	Ingested, chunks stored	Pass
DS-02	Upload a non-PDF	Rejected, 400	Pass
DS-03	Upload an empty file	Rejected, 400	Pass
DS-04	PDF yielding no text	422, document not retained	Pass
DS-05	Layout extraction fails	Ingest still succeeds	Pass
DS-06	Analysis queue unavailable	Document retained, warning logged	Pass
DS-07	Rename propagates to every chunk	All chunks updated	Pass
DS-08	Rename with a blank title	400, nothing written	Pass
DS-09	Rename an absent document	404	Pass
DS-10	Delete removes chunks, file and cached analysis	All removed	Pass
DS-11	Encrypt, then read chunk text	Ciphertext stored	Pass
DS-12	Decrypt with the wrong password	Refused, detected	Pass
DS-13	PDF route for an encrypted document	403	Pass
DS-14	PDF served inline, not as an attachment	Inline	Pass

2.3 Paper Writer

Table 2.3: Paper writer test cases

ID	Case	Expected	Result
PW-01	Compile a valid document	PDF produced	Pass
PW-02	Missing package declaration	Inserted, compiles	Pass
PW-03	Bare fragment with no preamble	Wrapped, compiles	Pass
PW-04	One malformed figure	Document still produced	Pass
PW-05	Index requested	.ind generated in Python	Pass
PW-06	Figure referenced from a project directory	Resolves	Pass
PW-07	Filename escaping the project directory	Refused	Pass
PW-08	Symbolic link out of the project	Refused	Pass
PW-09	Rename a project	Compile, preview and paths unaffected	Pass
PW-10	Cite a paper from the editor	Entry written, key returned	Pass
PW-11	Cite the same paper twice	One entry, same key	Pass
PW-12	A key already in the bibliography	Honoured, not recomputed	Pass
PW-13	Metadata improved after a citation	Original key retained	Pass
PW-14	An entry written by hand	Retained alongside	Pass
PW-15	A key written by hand collides	Suffixed, hand entry untouched	Pass
PW-16	Listing what can be cited	Nothing written to disk	Pass
PW-17	Citing a document not in the library	404	Pass
PW-18	Citations with no bibliography declared	Declaration supplied, references typeset	Pass
PW-19	Citations with no bibliography file	Nothing added, compile unaffected	Pass
PW-20	\citep, \citet, \nocite, page-qualified	All recognised	Pass
PW-21	A hand-written thebibliography	Respected, nothing added	Pass
PW-22	&, %, \$ in a title	Escaped, file valid	Pass

ID	Case	Expected	Result
PW-23	Capitalisation in a title	Preserved against the style	Pass
PW-24	Compile a cited document end to end	Numbered references and a reference list in the PDF	Pass

2.4 References and Correction

The two cases that open this table are the reason the section exists. Neither failure raises an error: one prints the wrong names, the other prints a question mark, and both look to the user like the feature not working.

Table 2.4: Reference storage and correction test cases

ID	Case	Expected	Result
RC-01	Eight authors stored and read back	Eight names, none split	Pass
RC-02	A name written surname first	One person, not two	Pass
RC-03	A list stored in the older format	Read as written	Pass
RC-04	A corrupted stored value	Read, not an error	Pass
RC-05	Correct title, authors and year together	One write to every passage	Pass
RC-06	Correct one field only	The others unchanged	Pass
RC-07	Correction with a blank title	Refused	Pass
RC-08	Correction with a blank year	Accepted	Pass
RC-09	A year that is not a year	Refused	Pass
RC-10	An author list of 200 names	Refused	Pass
RC-11	Correction to an absent document	404, nothing written	Pass
RC-12	Author line typed with commas	Split into names	Pass
RC-13	Author line typed with semicolons	Commas kept inside names	Pass
RC-14	Key from surname, year and title word	vaswani2017attention	Pass
RC-15	A paper with no usable metadata	Still citable	Pass

ID	Case	Expected	Result
RC-16	Two papers producing one key	Suffixed as BibTeX does	Pass

2.5 Capability, Models and Release

Table 2.5: Capability, model management and release test cases

ID	Case	Expected	Result
CM-01	Model sized against available, not installed, memory	Correct budget	Pass
CM-02	Prompt exceeding the loaded context	Map-reduce fallback	Pass
CM-03	Apple Silicon, no CUDA reported	GPU still used	Pass
CM-04	Vulkan device present, vendor tool absent	Advice still shown	Pass
CM-05	Task routed to an assigned model	Correct model	Pass
CM-06	Assigned model too large for free memory	Falls back, states why	Pass
CM-07	Version lower than published	Refused	Pass
CM-08	Release published as a draft	Detected and republished	Pass
CM-09	Update where the platform cannot prompt	Not installed	Pass

2.6 Interface

Table 2.6: Interface test cases

ID	Case	Expected	Result
UI-01	Press inside the page collapses the navigation	Collapses	Pass
UI-02	The collapse does not swallow the click causing it	Click delivered	Pass
UI-03	Reopening survives the next press	Stays open	Pass
UI-04	Automatic collapse resumes after navigating	Resumes	Pass
UI-05	Every feature resolves from its own path	All resolve	Pass

ID	Case	Expected	Result
UI-06	Nested path resolves to its parent feature	Resolves	Pass
UI-07	Unknown path	Falls back without error	Pass
UI-08	Workspace pages take the full window	Full width	Pass
UI-09	Update prompt with no confirmation available	Treated as not asked	Pass
UI-10	Pigments: earning records and applies	Correct	Pass

Chapter 3

TEST RESULTS

3.1 Suite Summary

Table 3.1: Automated suite results

Suite	Tests	Pass	Fail	Runtime
Backend (pytest)	1 085	1 084	0	~14 s
Interface (vittest)	222	222	0	~5 s
API contract	5	5	0	<1 s
Route reachability	16	16	0	<2 s
Total	1 328	1 327	0	

3.2 Metadata Extraction Accuracy

Two measurements, and the difference between them is the finding.

Table 3.2: Extraction accuracy: chosen versus randomly sampled papers

Field	Curated (18)	Random (56)
Title exactly correct	100%	92.9%
Author list exactly correct	100%	85.7%
Year exactly correct	100%	92.9%
All three simultaneously	100%	76.8%

The curated set was chosen by the authors. The random set was drawn across 15 arXiv subject classifications through the public interface, and scored against the archive’s own catalogue records, so neither the papers nor the expected answers were selected here.

The first random measurement was **62.5%** on all three fields. The whole difference was three typesetting conventions that had not been examined:

1. a physics template spacing a centred author line widely enough to be indistinguishable from column separation, so two authors read as five fragments and none as a name;
2. a template running from the affiliations into the abstract with no heading, so the search for authors continued into the body and collected section headings;
3. a margin threshold added to discard an archive stamp, which was also discarding the first line of any centred title wide enough to begin left of it.

Of the 13 remaining failures, **three are not extraction errors**: the page reads “M. GRENDÁR” and “Audrey Quessadda-Vial” where the archive record disagrees with the paper itself.

3.3 Ablation

To establish which rules carry the result, each was disabled and the corpus re-scored.

Table 3.3: Rule ablation on the curated set

Configuration	Exact	Failure mode
All rules	7/7	:
No column splitting	1/7	38 names lost entirely
No affiliation word list	4/7	Employers admitted; no name lost
No acronym rule	6/7	“Google AI Language” admitted

Splitting rows into cells is the load-bearing rule. The word list only ever admits extra rows and never loses a name, which is why it remains a tiebreaker rather than the mechanism.

3.4 Performance

Table 3.4: Ingestion profile, one 40 kB paper

Stage	Time	Share
Text extraction	6.0 ms	0.2%
Metadata extraction	1.1 ms	0.04%
Chunking	0.2 ms	0.01%
Embedding and search	3073.8 ms	99.7%

This measurement prevented a rewrite. Everything a systems-language rewrite would have touched is 0.3% of runtime, and the remaining 99.7% is already compiled native code.

Chapter 4

DEFECTS FOUND AND FIXED

4.1 By Severity

Table 4.1: Defects found during the testing phase

#	Defect	Severity	Status
D-01	Metadata wrong on every published paper tested	Critical	Fixed; 92.9% on unseen papers
D-02	Released version 1.0.0 did not start	Critical	Fixed; packaged validation added
D-03	Model fallback unreachable (guard tested for empty, failure was wrong)	High	Fixed; plausibility check
D-04	Summarisation prompt exceeded its own context window	High	Fixed
D-05	Apple Silicon pinned to CPU	High	Fixed
D-06	Release published as a draft; testers told they were current	High	Fixed; publish asserted
D-07	Graphics advice hidden from AMD/Intel/open-driver machines	High	Fixed
D-08	Sidebar collapse swallowed the click that caused it	Medium	Fixed
D-09	Update prompt read a missing confirmation as “declined”	Medium	Fixed
D-10	Interface panel displayed a model that served nothing	Medium	Fixed; reads the routing table
D-11	Two dashboard figures never displayed a value in any release	Medium	Fixed
D-12	Panes sized by a guessed constant; broke when a heading was removed	Medium	Fixed
D-13	Contract test validated a directory that was about to be deleted	Medium	Fixed; parameterised
D-14	Figure silently dropped while the compile reported success	Medium	Fixed; PDF inspected
D-15	Ligatures and accents reached the search index	Low	Fixed
D-16	Reader cannot open a PDF on Windows: the viewer’s worker module is served as <code>text/plain</code>	High	Fixed; MIME types declared
D-17	Gap finder reports the same gap more than once	Medium	Fixed; repeats merged

#	Defect	Severity	Status
D-18	Map nodes are difficult to select; navigation requires precision the pointer does not have	Medium	Fixed; 19 px target to 28 px
D-19	Reading surface is bright enough to tire the eye in a dim room	Low	Fixed; stock lowered to 89%

4.2 Closing the Cross-Platform Defects

All four defects from Section 5.1 were closed before this document was issued. Each is recorded here with its cause rather than only its symptom, because in three of the four the symptom named the wrong thing.

D-16, the reader on Windows. The message named a file, which made it look like a problem with the document, and the tester reasonably deleted the papers and ingested them again. No paper was involved. Python seeds its media type table from the Windows registry, which carries no entry for `.mjs`; the server therefore answered `text/plain`, and a browser refuses to execute a module script that is not served as JavaScript. The asset arrived and the browser declined it. The application now declares the six types a built interface is served as, before anything can answer a request. Invisible on Linux and macOS, where the built-in table is correct, which is why every suite passed while the reader was unusable on one platform in three.

D-17, repeated gaps. Nothing removed duplicates: each entry the model returned was given a fresh identifier, so two identical descriptions became two findings, drawn separately and counted separately. Repeats are now merged rather than discarded, since the second copy often carries evidence the first does not. The comparison is deliberately narrow, on case and spacing alone: a looser rule would eventually merge two findings that differ by one word, and the difference between evaluating on stationary and non-stationary data is the finding itself.

D-18, selecting a node. A paper is drawn as a point four to eight units across, and the click target was the drawing, which is six pixels once the map is scaled to fit. The drawing is unchanged, since larger discs would turn the plate into a bubble chart. What changed is that an invisible target carries the selection, and it holds a constant size on screen as the map zooms:

	Before	After
Visible point	5.9 px	5.9 px
Selectable area	5.9 px	28.0 px
Title selectable	no	yes
Papers visible at the default view	22	22

Opening the map already zoomed in was considered and rejected: any zoom high enough to help crops the map, and seeing the whole structure at once is what the view is for. The title was also inert, which mattered more than the target: it is roughly eighty pixels wide against a six pixel point, and a reader who has found the paper they want has almost always found it by reading the title.

D-19, the reading surface. The page was 94% luminance filling the screen. It is 89% now, and the text was pulled back from a contrast ratio of 15:1 to 11.9:1, since

near-black on cream is its own fatigue. An intermediate attempt lowered both the page and the surface beneath it and was rejected on sight as looking dusty: luminance fell while colour intensity rose, which reads as dirt rather than as paper. Every step of the text ramp, and both status colours, were measured against both surfaces and clear the AA threshold.

4.3 The Recurring Defect Class

Five of the defects above, D-01, D-03, D-06, D-07 and D-11, are the same mistake: *a check that tests for silence, when the failure mode is confident wrongness*. A guard asking whether a title is empty cannot catch a title that is wrong. A release asking whether a job failed cannot catch a job that succeeded into a draft. A dash meaning “nothing yet” cannot be distinguished from a dash meaning “the request failed”.

The correction is the same in each case: test for the shape of a right answer rather than for the absence of a wrong one.

Chapter 5

STATUS AND VERIFICATION

5.1 Cross-Platform Verification

Version 2.3.0 was published to the stable channel on 15 August 2026 and installed by testers on macOS and Windows. This is the first round in which the replaced interface and the citation feature were driven by people other than their authors, and it is the round that produced defects D-16 to D-19.

Table 5.1: What each platform was able to confirm

Platform	Result	Detail
macOS	Verified	Installed from the testing channel. A feature sweep found the application working, including writing, citing and compiling. Two defects were reported: the reading surface was tiring in low light (D-19), and the gap finder returned the same gap more than once (D-17). Both are fixed.
Windows	Verified, after one defect	Writing, citing and compiling were confirmed working. The reader could not display a PDF (D-16); the failure persisted after the affected papers were deleted and ingested again, which ruled out a corrupt document or a stale index and placed the fault in how the viewer’s worker module is served. Fixed.
Linux	Verified	The development platform. Every automated suite runs here, and the ingest, map, write, cite and compile paths are exercised by hand continuously, including against 2.3.0. The packaged installer has been installed and run.

Two results are worth stating separately from the table.

Scribe was confirmed working on both macOS and Windows, including writing and compiling. That closes the largest of the three items this document previously listed as outstanding, and it is the item that carried the most risk, since the writing path depends on a typesetting engine bundled inside the application and on filesystem behaviour that differs between platforms.

Map navigation was raised independently on both platforms (D-18), and is fixed. When two testers on different operating systems describe the same difficulty without prompting, it is a property of the design rather than of a machine, and it is recorded here as a usability defect rather than as feedback.

5.2 A Defect No Test Environment Could Have Caught

One defect from the citation work is recorded for the method rather than the cause, because it changes what the interface suite can be relied on to prove.

The citation list did not appear. Every automated test passed, including tests that type into an editor and assert that the list opens with the expected rows, because those tests run in a simulated document object model that computes no layout: every element it reports is at the origin and has no size. Driving the real application and reading the element's position back showed the list present, populated with all twenty papers, and drawn 1382 pixels down a window 1000 pixels tall, inside a region that hides rather than scrolls what exceeds it. A second cause was found the same way: the compiled document displayed beside the editor is drawn above the list and concealed the column holding the citation keys.

The consequence for this document is narrow and should not be overstated. The interface suite proves that components mount, that state changes as expected, and that the calls they make exist. It cannot prove that anything is visible. Screenshots taken from the running application, and the platform verification in Section 5.3, are the only evidence of that, which is part of why the outstanding items below are treated as blocking rather than cosmetic.

5.3 Verification Status

Every feature described in the project report is implemented, and every defect found during the testing phase is fixed. The application is verified on all three platforms it ships to.

All automated suites pass on Linux, which is the development platform: 1 328 tests across four layers, of which one is skipped rather than run. That test asks the running server what media type it gives a JavaScript module, which is the fix for D-16; it needs a built interface to serve, and skips rather than pretending to pass when there is none. The application has been released publicly through sixteen versions, and version 2.3.0 has been installed and exercised by testers on macOS and Windows as recorded in Section 5.1. Four defects came out of that round, D-16 to D-19. None is a regression introduced by the release: all four were present in earlier builds without having been observed. All four are now fixed, as recorded in Section 4.2.

Table 5.2: Verification by platform

Platform	Status
Linux	Verified. The development platform: every automated suite runs here, and the ingest, search, map, gap, write, cite and compile paths are exercised by hand continuously, including against 2.3.0. The packaged installer has been installed and run.
macOS	Verified by a tester on 2.3.0. Writing, citation and compilation confirmed working, including the citation list drawn at the cursor. The two defects found here, D-17 and D-19, are fixed.

Platform	Status
Windows	Verified by a tester on 2.3.0. Writing, citation and compilation confirmed working. The defect found here, D-16, is fixed, together with D-18 which was reported from both platforms.

Two properties of the fixes are worth stating, because they are the reason this section can be short. Each of D-16 to D-19 is covered by an automated test that fails without the fix, so a regression is caught by the suite rather than by a person. And D-16, which is the only one whose cause is specific to a single operating system, is tested by asking the running server directly what it says a JavaScript module is, rather than by reasoning about what it ought to say.

What remains is not verification but capability: the models trained for this application, the layout classifier for metadata extraction, and the editor work set out in the project report's Future Scope. Those are the subject of the next phase of the project rather than gaps in this one.

5.4 Revision

This document will be revised as the defects in Section 4.1 are closed and the verification in Section 5.3 is completed. The suite figures in Chapter 3 are current as of the date of issue and are reproducible by running the suites named beside them.